

EAS 6134 Inverse Methods in EAS
Homework 5: A Linear Inverse Problem

For background read the paper by Constable et al. (2018, Geophysical Journal International. 215,49-60, doi: 10.1093/gji/ggy263). The data set you are using is similar to that in the paper, but there are some differences.

1. Two electric field datasets are uploaded on the canvas website (Files->Homework 5). The file zline.txt is a set of electric field measurements along a line expressed as northing (distance along track), vertical electric field, error and the second file zmap.txt provides data as a function of 2D position, in the form easting, northing, vertical electric field, error.

The units are meters and microvolts per meter. These are 30 second averages during AUV flights. The errors are stacking errors (What does this mean?) and are probably too small.

We want to make a prediction of the field at positions when data are unavailable and will do so by constructing a series of possible models describing the field variations [i.e., recreating Section 5.1 in Constable et al. (2018)]. Let's start with the data in zline.txt with the goal of providing a representation of electric field at any point between the available measurements. This is the simplest forward problem of all where you might think of the data themselves as the forward problem.

Start with the idea of drastic simplification and consider a simple discretized overdetermined least squares (ODLS) parameter estimation problem, i.e one in which the discretization is sufficiently coarse that you have more observations than model parameters.

(a) Set up and solve the ODLS problem. Plot the model and residuals. Calculate the model norm and the norm of the residuals.

(b) Now recast the problem as a linear inverse problem, with many more parameters than observations. Without additional constraints this will be an underdetermined problem for which there is not a unique solution. Choose a suitable norm and find the *norm minimizing model* that fits the observations exactly — this is a form of interpolation, and can be done either analytically or numerically. Numerically is easier and provides a potentially more general framework for future problems you might want to solve. Evaluate the model norm and compare with the results of part (a).

(c) Proceed to the more realistic modeling approach where the data are not fit exactly but instead fit to a target misfit level. Explain how you might choose a suitable target misfit. Evaluate and plot a series of models with various misfit levels. Tabulate values of the Lagrange multiplier, rms misfit, and model norm from your work. Plot the model predictions and residuals from the data.

2. Now let's proceed to the grid of values provided in zmap.txt and solve the inverse problem of constructing a 2D map of electric field as a function of position. You are essentially attempting to recreate Figure 5 in Constable et al. (2018). You might want to modify your forward model slightly. Plot the model norm as a function of rms misfit and also as a function of Lagrange multiplier.

For this exercise, it is strongly advised that you avoid the use of a global basis such as polynomials, Hermite functions, Fourier series, etc. Instead use a local basis such as a piecewise constant or piecewise linear function with numerous parameters, allowing implementation of an approximate first derivative smoothness constraint.